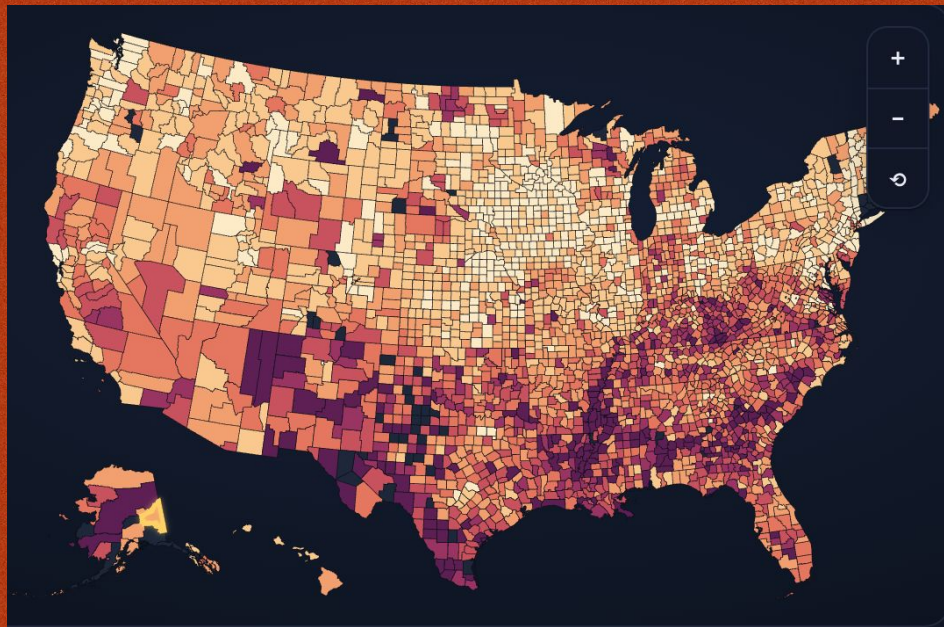


Can You Fix It?

Understanding stressors & considering potential fixes



Big Question

If a school's surrounding conditions are contributing to a high score on these indexes, what might it cost to improve them?

Economic Stress	Unemployment, Poverty, Access to broadband internet, Single-parent households
Education Stress	Less than HS, 2-year college or higher, and Linguistic isolation
Health Stress	Access to healthcare, Infant mortality rate, SNAP recipients, Low birth weight, and Lead exposure risk
Housing Stress	Housing vacancy rate, Housing affordability, and Park access
Crime Stress	Violent crime rate and Incarceration rate

Our Data

- Open Data Index for Schools (ODIS)
- 23,599 high schools
- Previously defined indexes – composite stress
- Stress breakdowns
- Helpful for policy makers & students

Can You Fix It? School Stress Intervention Calculator

Select State:

NC

Select School:

Carrboro High

Current School Profile

Indicator	Current.Value	National.Percentile
Composite Stress	17.00	NA
Park Access	21.00	28.00
Broadband	9.00	12.00
Healthcare	3.00	30.00
Violent Crime	14.00	23.00
SNAP Recipients	43.00	38.00
Poverty	14.00	3.00

Our Methodology

1. Gather the data. ODIS—For each high school, Community composite stress score broken down with interventions

3. Translate indicator changes into overall stress. Using a regression, we calculate how much lowering one indicator (like broadband or crime) would lower the school's overall composite stress score.

2. Set a realistic goal for each intervention indicator. We compare each school to the best-off 10% of schools nationally. If a school is already there on an indicator, we leave that indicator alone. If not, the gap between where it is and that benchmark is the most it can improve.

4. Put a price on it. We start with a national average cost per point for each intervention, then adjust it up or down: bigger counties cost more, and states with higher prices cost more.

Determining Baseline Cost

Each base cost comes from a published program matches to what the indicator measures.

- Park access: \$400,000/pt (new neighborhood park; Trust for Public Land data)
- Broadband: \$120,000/pt (internet subsidy; Affordable Connectivity Program)
- Healthcare: \$320,000/pt (insurance enrollment help; parent mentor trial)
- Violent crime: \$90,000/pt (Healthy Communities for Youth cost study)
- SNAP: \$700,000/pt (income support; outreach would raise the score)

*Our calculator uses national-average base costs so every school can be compared on the same terms. Real costs depend on local land prices, existing infrastructure, program design and who delivers it. A school district or city can replace any base cost with local figures to get a community-specific estimate.

Weights vs. Regression - An OLS on 10,000 Schools

$$\text{Composite Stress} = \beta_0 + 0.297 \cdot \text{Broadband} + 0.180 \cdot \text{Healthcare} + 0.141 \cdot \text{SNAP} + 0.122 \cdot \text{ViolentCrime} + 0.042 \cdot \text{ParkAccess}$$

$$R^2=0.865$$

	Formula weights (done by ODIS)	Regression
Park access	$1/15 = 0.067$	0.042
Broadband	0.05	0.297
Healthcare	0.04	0.180
Violent crime	0.10	0.122
SNAP	0.04	0.141

Each CAP is at the difference from the school's current value to the **national 10th percentile target**

Example: Broadband matters $\sim 7\times$ more than ODIS's formula assumes

COST model

What do points mean: Each indicator is an ODIS stress score from 0 to 100 (higher = worse conditions). "Lowering the broadband score by 5 points" means moving 5 points toward the best-connected communities.

01 POPULATION MULTIPLIER

$\sqrt{\text{county_pop} / 85,000}$, capped between 0.5x and 3.0x, then multiplied by the state's BEA Regional Price Parity (national avg = 1.0). Larger counties and higher-cost states cost more per point.

02 INTERVENTION CAPS

Each lever can only lower an indicator down to its national 10th-percentile stress score (the targets at right). No school is modeled as doing better than the top 10% of US schools already do. The composite score also can't drop below 0.

03 GREEDY ALLOCATION

No single intervention can supply more than 40% of the target.

NATIONAL PERCENTILE TARGETS

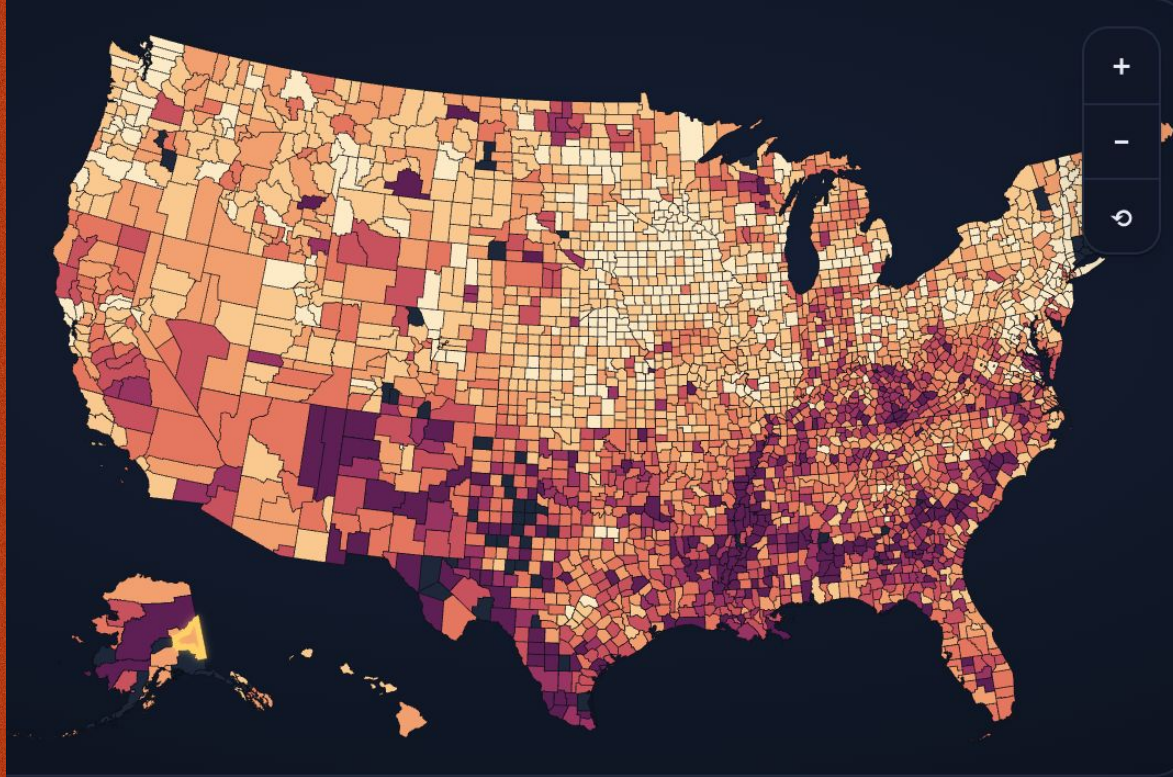
lower-is-better Each stress score is lowered towards the best 10% of schools.

Eligibility: a lever is recommended whenever the indicator is above its 10th-percentile target. If it's already at or below the target, there's nothing left to improve. SNAP also requires a poverty score above 30.

Indicator Intervention	Target
Broadband access gap	9
Healthcare access gap	2
Park access gap	4
Violent crime exposure	8
SNAP dependency	31

*Actual percentiles of the stress score distributions across all schools in data set

Stress Visualization



Light = Low Stress
Dark = High Stress

input

Recommended Optimal Plan

This is the cheapest combination of interventions to reduce stress by your target.

Target stress reduction (points):

5

Intervention	Improve.by	Stress.reduced	Cost
crime	6.00	1.11	666,771
park	16.20	2.00	8,001,252
healthcare	1.00	0.07	395,124

Total cost: \$9,063,147

Original score: 17

New score: 12

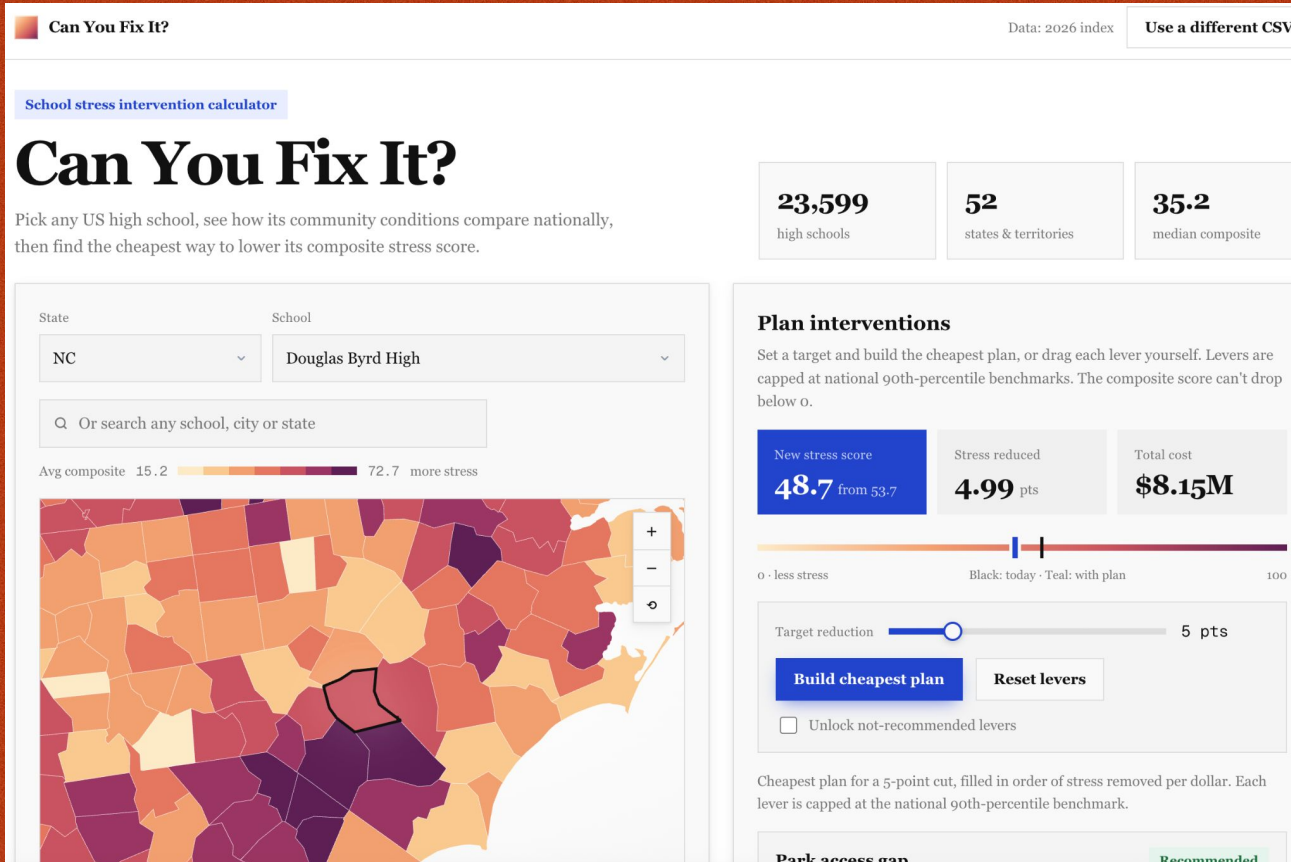
Pop. multiplier: 1.23x

output

Each input below is capped at the maximum realistic improvement for **Carrboro High**.

- *Broadband* can improve by at most 0 points, and
- *Crime* can be reduced by at most 6 points.
- Costs are adjusted by a 1.23x population multiplier based on 147,000 residents in this county.

Demo



Important Caveats!

1. Correlation, not necessarily causation
2. Regression gives an optimistic estimate
3. Coefficients aren't fully specific
4. Indicators interact!

Applications

Going Deeper

Expand analysis to be more tailored toward smaller regions (eg. costs)

For students

Explore which local factors shape their own community's stress score.

For policymakers

Estimate realistic intervention costs and see which levers cut stress fastest per dollar for a given school.

NOTE: Not *dynamic yet*- it's a static snapshot, but the results refresh when a new ODIS CSV is loaded.

THANKS

Any Questions?

CREDITS: This presentation template was created by **Slidesgo**, and includes icons, infographics & images by **Freepik**

Please keep this slide for attribution

CITATIONS

Hawken, A., Minar, N., Choudhary, R., & Kulick, J. (2026). Open Data Index for Schools (ODIS) (Version 1) [Data set]. Johns Hopkins Research Data Repository. <https://doi.org/10.7281/T170WN53>. CC BY 4.0.

U.S. Census Bureau. (2024). American Community Survey 5-Year Estimates, 2019–2023, Table B01003: Total Population [Data set]. <https://data.census.gov>. Accessed via tidycensus (Walker & Herman), <https://CRAN.R-project.org/package=tidycensus>

U.S. Bureau of Economic Analysis. (2023). Regional Price Parities by State, 2022 (Table SARPP) [Data set]. <https://www.bea.gov/data/prices-inflation/regional-price-parities-state-and-metro-area>

U.S. Census Bureau. (2010). Cartographic Boundary Files: Counties [Data set]. Accessed via Plotly, <https://github.com/plotly/datasets>

Presentation template by Slidesgo, including icons, infographics and images by Freepik.

BASE COST CITATIONS

Trust for Public Land. (2026). *The Undeniable ROI of Community Parks and Green Spaces*. <https://www.tpl.org/parks-undeniable-roi-report>

Trust for Public Land. (2021). *Community Schoolyards: Our Schoolyard Pledge*. <https://www.tpl.org/resource/community-schoolyards-report-2021>

Federal Communications Commission. *Affordable Connectivity Program* (up to \$30/month per household). <https://www.fcc.gov/acp>

Kane, J. (2025, September 9). *BEAD's Benefit of the Bargain Round So Far: The Good, the Bad, and the Uncertain*. Information Technology and Innovation Foundation. <https://itif.org/publications/2025/09/09/beads-benefit-of-the-bargain-round-so-far/>

Flores, G., Lin, H., Walker, C., Lee, M., Currie, J. M., Allgeyer, R., Fierro, M., Henry, M., Portillo, A., & Massey, K. (2016). Parent mentors and insuring uninsured children: A randomized controlled trial. *Pediatrics*, 137(4), e20153519. <https://doi.org/10.1542/peds.2015-3519>

Pan, Z., Chapman, D. A., Sullivan, T. N., Bishop, D. L., & Kimmel, A. D. (2024). Healthy Communities for Youth: A cost analysis of a community-level program to prevent youth violence. *Prevention Science*, 25(7), 1133–1142. <https://doi.org/10.1007/s11121-024-01729-z>

Shafer, P. R., Gutiérrez, K. M., Ettinger de Cuba, S., Bovell-Ammon, A., & Raifman, J. (2022). Association of the implementation of Child Tax Credit advance payments with food insufficiency in US households. *JAMA Network Open*, 5(1), e2143296. <https://doi.org/10.1001/jamanetworkopen.2021.43296>